There Is No Single LoCoMo Score:
A Standardized, Vendor-Audited Re-evaluation of LLM Memory Systems

Kamal Gurbanov
PersonalMem Project · Independent
kamalgurbanov916@gmail.com

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NOT FOR SUBMISSION without filling PENDING cells.

Confidence legend: **[V]** verified (primary source) **[P]** partially/secondary-verified **[U]** unverified (primary not located)

Abstract

Long-term conversational memory systems for large language models report wildly divergent accuracies on LoCoMo, the de-facto benchmark for very-long-term dialog memory: Zep ∼84% (later corrected to 58.44%, then counter-claimed at 75.14%), Mem0 67.13% and subsequently ∼92.5%, with each vendor publicly disputing the others' methodology. We show that these numbers do not diverge because any party is lying. **There is no single "LoCoMo score."** Each headline number is the output of a *different protocol*, run by an *interested party on a competitor's configuration*, and several widely-cited head-to-head comparisons (e.g., "Zep 63.8 vs. Mem0 49.0") are *cross-source splices* that were never produced under one harness. We make four contributions.

First, a **sourced forensic decomposition** of the dispute into four nameable methodological forks — (A) denominator/adversarial-category handling, (B) who-configured-the-loser, (C) judge- and answer-model drift, (D) benchmark-identity confusion — each traced to a primary URL. Second, a **provenance correction**: the famous "84%" does *not* appear in Zep's arXiv paper [Zep Team, 2025a] (which evaluates only DMR and LongMemEval and contains no LoCoMo evaluation); it originates in Zep's separate evaluation repository and blog. Third, **one frozen, fully-specified, auditable protocol** that neutralizes all four forks and is now live in the harness: every system retrieves context, a single **pinned answer model** (Claude Sonnet) *generates* the answer from (question + retrieved context), and a single **pinned LLM judge** (Claude Sonnet) scores generated-answer-vs-gold correctness — the credible, vendor-comparable **judge-accuracy** J. Fourth, a **keyless real-data floor** computed on the real `snap-research/locomo` corpus at \$0.00 — bm25 token-F1 0.0596 and tf-idf 0.0520 over N=1540 answerable questions — plus a **free-tier single-persona pilot** (p01, 40 Q, 1 seed, \$7.74) that yielded longcontext J=**0.775** and keyless floor J=0.025. Subsequent free-tier vendor pilots (same persona, polling ingest) measured all three vendors — **Mem0** J=0.475, **Letta** J=0.200, **Zep** J=0.100 — each landing between the long-context control and the retrieval floor; only the **full 5-seed/10-persona run remains PENDING** (see §7.2).

1 Introduction *[DRAFTED]*

Personal AI assistants are judged increasingly on memory: the ability to recall, across months of conversation, what a user said, when they said it, and how it relates to everything else. The benchmark that has come to stand in for this capability is **LoCoMo** [Maharana et al., 2024], ten multi-session conversations of roughly 300 turns and 9K tokens each, with questions tagged into single-hop, multi-hop, open-domain, temporal, and adversarial categories. Commercial memory vendors compete openly on their LoCoMo accuracy. The trouble is that the reported accuracies are mutually irreconcilable.

The public record reads as a dispute. Zep's evaluation artifact reported roughly **84%**. Mem0 re-ran Zep and reported **58.44%** $\pm$ **0.20**, attributing the gap to an inconsistent denominator. Zep rebutted with **75.14%** $\pm$ **0.17**, attributing that gap to Mem0 having mis-configured Zep's adapter. Mem0's own paper reports **67.13%**, and a later Mem0 announcement reports $\sim$**92.5%**. Independent systems land elsewhere again — Memobase 75.78%, MemMachine 91.69%. A reader trying to answer the simple question "which memory system is best on LoCoMo?" finds no answer, only a crossfire.

Our claim is that this crossfire is **not a disagreement about architecture** — it is a disagreement about *measurement that no one fixed*. There is no single LoCoMo score because every number above was produced under a different protocol, and in several cases by the *competitor* of the system being scored. Once the protocols are named and held constant, the 84/58.44/75.14/92.5 spread resolves into **four methodological forks**, not four architecture verdicts.

We do four things. (1) We **forensically decompose** the dispute, tracing every number to a primary source with an explicit confidence flag. (2) We **correct the provenance** of the most-cited number, the "84%," which is widely but wrongly attributed to Zep's arXiv paper. (3) We specify **one frozen, fair, fully-reproducible protocol** that neutralizes all four forks and run it on the real LoCoMo data, with a pre-disclosure window so each vendor sees its own number before publication. (4) We report a **keyless real-data baseline floor** now, at zero cost, and present the vendor results table as **pending the budget-gated keyed run**.

2 Background & Related Work *[DRAFTED]*

LoCoMo. Maharana et al. [2024] (ACL 2024; `snap-research/locomo`; CC BY-NC 4.0; arXiv:2402.17753). Ten multi-session conversations between two speakers, $\sim$300 turns and $\sim$9K tokens each, with question-answer pairs tagged by category. The repository's official primary metric is SQuAD token-overlap F1. Standard practice evaluates Categories 1–4 (single-hop, multi-hop, open-domain/commonsense, temporal) and excludes Category 5 (adversarial), whose ground-truth answers are not usable under the LLM-judge protocol. **[V]** (Mem0 paper, §eval).

Mem0. Mem0 Team [2025a] (arXiv:2504.19413). A memory layer that extracts and consolidates salient facts. Its paper reports LoCoMo under a `gpt-4o-mini` answer-and-judge regime, Categories 1–4 only, with ±std over multiple seeds — the most fully-specified vendor protocol in the dispute, and the methodological reference point for this audit. Mem0 overall $J = 67.13 \pm 0.65$; graph variant 65.71 ± 0.45; re-runs Zep at 65.99 ± 0.16. **[V]**

Zep. Zep Team [2025a] (arXiv:2501.13956, "A Temporal Knowledge Graph Architecture for Agent Memory"). **Critically: Zep's paper evaluates only two benchmarks — DMR (94.8% `gpt-4-turbo`/98.2% `gpt-4o-mini`) and LongMemEval (63.8 `gpt-4o-mini`/71.2 `gpt-4o`) — and contains no LoCoMo evaluation and no "84%" figure.** **[V]** The LoCoMo numbers attributed to Zep originate in a separate artifact (the `getzep/zep-papers` `locomo_eval` code and an accompanying blog post).

Memora/FAMA. Uddin et al. [2026] (arXiv:2604.20006). Forgetting-aware memory accuracy. Cited as adopted prior art for the project's Track-2 metric work (`eval/metrics.fama`); not a system tabulated against LoCoMo here.

Adjacent benchmarks. Wu et al. [2024] (LongMemEval, arXiv:2410.10813) and DMR are different datasets with different protocols. A recurring confusion (§4, Fork D) is quoting a LongMemEval or DMR number as though it were a LoCoMo number. We treat *one benchmark per table* as a hard rule.

3 The Dispute *[DRAFTED]*

Lead finding (provenance correction). The famous "Zep claimed 84% on LoCoMo" did **not** come from Zep's arXiv paper. [V] The Zep paper [Zep Team, 2025a] evaluates Zep on **only DMR and LongMemEval** and contains **no LoCoMo evaluation and no 84% figure at all.** The 84% originates from a separate Zep benchmarking artifact — the `getzep/zep-papers` `locomo_eval` evaluation code and an accompanying blog — which is precisely what Mem0's GitHub issue #5 [Mem0 Team, 2025b] corrects. We attribute the 84% to the **eval repo/blog**, not the paper.

4 Why the Numbers Diverge: Four Methodological Forks *[DRAFTED]*

Fork A — Denominator / adversarial-category handling (explains 84 → 58.44, the largest gap). [V] Per Mem0's issue #5 [Mem0 Team, 2025b], Zep's 84% counted Category-5 (adversarial) correct answers in the **numerator** while the **denominator excluded** Category 5 — an inconsistent normalization that inflates the score by ∼25.56 points. Re-scoring Zep on a consistent Categories-1–4-only basis yields **58.44%** ± **0.20**. This is an **arithmetic/normalization fork, not an architecture fork.**

Status: Mem0's description of the inflation is verified; we have **not** independently re-executed Zep's original script to confirm the exact arithmetic — that re-execution is precisely the audit's no-spend reproduction target (§8, honesty ledger).

Fork B — Who configured the loser (explains 58.44 ↔ 75.14, the same system). Every headline number is a party scoring its competitor's system. [V] Zep's rebuttal [Zep Team, 2025c] claims Mem0's harness mis-set Zep three ways: (i) used a single-user graph but assigned *both* speakers the user role; (ii) passed timestamps by appending to message text instead of Zep's dedicated `created_at` field (specifically hurts the temporal category); (iii) ran searches sequentially rather than in parallel (inflates *latency*, not accuracy). Fixing (i)–(ii), Zep reports **75.14%** ± **0.17**. So 58.44 and 75.14 are the *same* Zep system under hostile vs. friendly configuration — neither is "the" Zep score.

Fork C — Judge and answer-model drift (silently shifts every number). LoCoMo's J is LLM-as-judge, so the judge and answer model are part of the measurement. [V] Mem0: *"All language model operations utilized gpt-4o-mini."* [V] Memobase judged with `gpt-4o`. [P] MemMachine answered with `gpt-4.1-mini` and reports 91.69. [U] Mem0's new 92.5 and Zep's 75.14 do **not** state their judge/answer model. Both stronger answer models and more lenient judges raise J. The ∼92.5/91.69 cluster (newer, stronger models) versus the ∼58–75 cluster (`gpt-4o-mini` era) is therefore **largely a model-generation effect, not purely an architecture win.**

Table 1: Claims-as-published (provenance-annotated; **NOT** single-harness results).

#	Claim	Claimed by	Benchmark	Answer model	Judge	Cat-5 in denom?	Configured by	Conf
1	Zep ∼84%	Zep (eval repo/blog) [Zep Team, 2025b]	LoCoMo	unspec.	unspec.	yes (inflated)	Zep (self)	[P]
2	Zep 58.44% ± 0.20	Mem0 (re-running Zep) [Mem0 Team, 2025b]	LoCoMo	gpt-4o-mini	Mem0	no (corrected)	Mem0	[V]
3	Zep 65.99% ± 0.16	Mem0 paper T1 [Mem0 Team, 2025a]	LoCoMo	gpt-4o-mini	gpt-4o-mini	no	Mem0	[V]
4	Zep 75.14% ± 0.17	Zep rebuttal [Zep Team, 2025c]	LoCoMo	unspec.	unspec.	no	Zep (self)	[V]
5	Mem0 67.13 ± 0.65; Mem0^g 65.71 ± 0.45	Mem0 paper T1 [Mem0 Team, 2025a]	LoCoMo	gpt-4o-mini	gpt-4o-mini	no	Mem0 (self)	[V]
6	Mem0 92.5 (new algo)	Mem0 [Mem0 Team, 2026]	LoCoMo	unspec.	unspec.	unstated	Mem0 (self)	[V] num., [U] prot.
7	Mem0 94.4	Mem0 [Mem0 Team, 2026]	LongMemEval	unspec.	unspec.	n/a	Mem0 (self)	[V] num., [U] prot.
8	Zep 63.8/71.2	Zep paper [Zep Team, 2025a]	LongMemEval	gpt-4o-mini/gpt-4o	LME	n/a	Zep (self)	[V]
9	"Zep 63.8 vs. Mem0 49.0 (GPT-4o)"	Secondary aggregators	LongMemEval	mixed	mixed	n/a	not one harness	[P]/[U] (splice, §4D)
10	Zep 94.8/MemGPT 93.4	Zep paper [Zep Team, 2025a]	**DMR** (not LoCoMo)	gpt-4-turbo	DMR	n/a	Zep	[V]
—	Memobase 75.78; temporal 85.05	Memobase [Memobase Team, 2025]	LoCoMo	unspec.	gpt-4o	no	Memobase	[V]
—	MemMachine 91.69	MemMachine [MemMachine Team, 2025]	LoCoMo	gpt-4.1-mini	unspec.	no	MemMachine	[P]

Per-category Mem0-paper cells [V]: Mem0 overall 67.13; multi-hop 51.15 ± 0.31; open-domain 72.93 ± 0.11; temporal 55.51 ± 0.34. Zep in same table: single-hop 61.70 ± 0.32, multi-hop 41.35 ± 0.48, open-domain 76.60 ± 0.13, temporal 49.31 ± 0.50. ***Caveat****: an automated extractor mislabeled the Mem0 67.13 row as "single-hop"; 67.13 is the overall J — re-confirm cell layout against the PDF before final typesetting (§8, honesty ledger).*

Fork D — Benchmark-identity confusion (the "63.8 vs. 49.0" splice). [V] The "Zep 63.8 vs. Mem0 49.0 on LongMemEval (GPT-4o)" pairing that circulates in aggregators is a **cross-source splice**: 63.8 is Zep's own paper number for Zep + `gpt-4o-mini` (Zep + `gpt-4o` is 71.2), while the 49.0 for Mem0 comes from a third party (Vectorize), not a shared head-to-head harness. Separately, Zep's DMR 94.8% is routinely quoted *as if* it were a LoCoMo number. **Any Zep-vs-Mem0 pairing not produced by a single run is unverified.** [U] for the 49.0 specifically.

Table 2: Fork → mitigation traceability.

Fork	Symptom	Protocol kill-switch
A Denominator / adversarial	84 → 58.44 (+25.56 pts)	Cat 1–4 only; Cat-5 out of *both* numerator and denominator; per-category denominators published
B Who configured the loser	58.44 ↔ 75.14 (same system)	Vendor-blessed/sign-off configs; all configs published verbatim
C Judge/answer-model drift	58–75 vs. 92.5 clusters	One pinned answer + judge model; second judge for robustness; unstated-model cells flagged
D Benchmark-identity confusion	"63.8 vs. 49.0" splice; DMR quoted as LoCoMo	One benchmark per table; splices shown only as provenance-annotated claims

5 A Standardized, Auditable Protocol (Methods) *[DRAFTED]*

We re-run **all** memory systems on the **real** `snap-research/locomo` data (commit `3eb6f2c`, CC BY-NC 4.0; 10 conversations, 5882 turn-Documents) under a **single frozen harness** (`eval/runner.py` + `eval/metrics.py`). Each design choice maps to the fork it neutralizes.

5.1 Frozen dataset (kills Fork A)

Source `snap-research/locomo` @ `3eb6f2c`, data file `external/locomo/data/locomo10.json`. **Scored set: Categories 1–4 only,** N=1540 (single_hop 841, temporal 321, multi_hop 282, open_domain 96). Category 5 (446 adversarial) is **EXCLUDED from both numerator and denominator** of the headline and reported separately as an abstention diagnostic. Per-category denominators are published verbatim.

5.2 The generate-then-judge pipeline (the measurement, as implemented)

This is the core of the protocol and is live in the harness (`eval/answer_gen.py`, `eval/judge.py`, wired into `eval/runner.py` via `-answer-mode generate -judge llm`). For each (system, question, seed) cell:

1. **Retrieve.** The system under test returns its retrieved context in a uniform field `QueryResult.extra["retrieve` (a `list[str]`): top-k passages for `bm25/tfidf_rag`, surfaced memories for `mem0`, graph facts + node summaries for `zep`, and the stuffed persona corpus for the `longcontext` control. The retrieval step is the *only* part that differs across systems.
2. **Generate (pinned answer model).** A single pinned answer model — `claude-sonnet-4-5` (temperature 0, `max_tokens` 128) — generates a concise answer grounded **only** in that retrieved context. If context is empty the model abstains ("I don't know") rather than hallucinate.

3. **Judge (pinned LLM judge).** A single pinned judge model — **`claude-sonnet-4-5`** (temperature 0) — scores the generated answer against gold as a binary correct/incorrect in the LoCoMo/Memora style: tolerant of paraphrase, formatting, and equivalent dates/units; strict about the actual fact. An empty prediction short-circuits to *incorrect* with **no paid call**.

J (judge-accuracy) is the mean of these correct verdicts and is **the headline, vendor-comparable metric**. We also report **token_f1** (SQuAD token-overlap F1, the LoCoMo-official metric) and **exact_match**. ≥ 5 seeds per (system, dataset) cell, reporting **mean ± std**. A temporal per-category breakdown is always reported; temporal is the fork-sensitive category (Fork B-ii).

Verification (no vendor spend). The full pipeline was exercised end-to-end on a real-LoCoMo slice with bm25 retrieval and a hard \$0.60 cap: it produced J and token_f1, billed real cost \$0.0086, and the budget gate halted cleanly at 90% on a \$0.006 re-run. The motivating case: gold "*self-care is important*," bm25-grounded generation "*Melanie realized that self-care is really important,*" token_f1 = 0.6 but **judge = correct** — the judge credits the fact the token metric penalizes.

5.3 Pinned models (kills Fork C)

Answer model and primary judge: **`claude-sonnet-4-5`** (the one API key the PI holds). A **robustness judge** (second model; `gpt-4o-mini` if a second key is authorized) is reported alongside. **Disclosure rule:** every table states its answer and judge model; where a vendor's number used an unstated model (Mem0's 92.5, Zep's 75.14, both [**U**]), that cell is marked "model-unstated (vendor-reported)" and is **not** presented as comparable.

5.4 Frozen system/config spec (kills Forks B and D)

Systems: keyless real-data floor (`bm25`, `tfidf_rag`, no spend, runnable now, reported as a *retrieval floor*); long-context control (`longcontext/claude-sonnet-4-5`); vendor systems (`mem0`, `zep`, `letta`; PI-gated keyed run). **Vendor-config sign-off:** use each vendor's own published/blessed adapter config as default; publish all configs verbatim, including Zep's three contested settings (single- vs. dual-user role, `created_at` vs. appended timestamp, sequential vs. parallel search). **One benchmark per table:** a LoCoMo cell never shares a table with LongMemEval or DMR.

5.5 Category-5 handling

Excluded from the headline (Fork A); systems that emit abstentions are scored on Cat-5 with an abstention/over-answer diagnostic in its own table (Table 5), never folded into the J/token_f1 headline.

5.6 Pre-disclosure / fair-play protocol

Before any public release: (1) freeze the protocol and run it; (2) notify each vendor (Mem0, Zep, Letta) with their adapter config, their number(s), the exact reproduce command, and the git SHA + dataset commit; (3) open a **comment window** (∼10 business days) for the vendor to flag a misconfiguration or supply a blessed config; (4) publish protocol, all configs, all seeds, the manifest, and per-cell results.

5.7 Reproducibility contract

Every run launches from a frozen config; every cell carries a seed; the runner writes a manifest (git SHA, configs, seeds, dataset commit) to `results/<tag>.json`; runs are **budget-gated** (hard USD ceiling, halt at 90%) and **resumable** (same `-tag` continues without re-spending completed cells).

6 Experimental Setup *[DRAFTED]*

Harness: `eval/runner.py` (budget-gated, resumable, manifest-emitting), driving the generate-then-judge pipeline of `eval/answer_gen.py` + `eval/judge.py` under `-answer-mode generate -judge llm`. Metrics: `eval/metrics.py`. Dataset commit `3eb6f2c`; corpus 5882 turn-Documents at `corpus/locomo_p01..p10/documents.jsonl`. Pinned answer + judge model `claude-sonnet-4-5`; ≥5 seeds per cell; Sonnet-class robustness judge (per §5). License compliance: raw CC BY-NC data is not redistributed; clone + commit-pin instructions are provided instead.

7 Results *[PENDING-DATA]*

7.1 Keyless real-data baseline floor — DONE, $0.00

We loaded the real LoCoMo corpus and ran `bm25` and `tfidf_rag` over the N=1540 answerable (Cat 1–4) questions: **3080 cells scored, cost $0.00.** These extractive baselines return the best query-overlapping sentence span of the top-ranked conversation turn with no LLM generation; LoCoMo gold answers are short generated phrases, so EM is exactly 0 and token-F1 is a low partial-overlap **retrieval floor** — **NOT comparable** to Mem0/Zep's generation-based accuracies (∼58–92%).

Table 3: Keyless retrieval floor (real LoCoMo, Cat 1–4, N=1540, $0.00, seed=1).

System	EM	token_f1	single_hop	open_domain	multi_hop	temporal
bm25	0.0000	**0.0596**	0.0898	0.0244	0.0291	0.0179
tfidf_rag	0.0000	**0.0520**	0.0783	0.0279	0.0242	0.0146
combined	0.0000	0.0558	—	—	—	—

Per-category N: single_hop 841, temporal 321, multi_hop 282, open_domain 96. These are a retrieval floor, not system scores.

7.2 Vendor + long-context results — pilot run; vendor cells blocked on free tiers

The main vendor comparison is structured around J **(judge-accuracy)**, produced by the generate-then-judge pipeline of §5.2. A **free-tier, single-persona (p01) pilot** (40 questions, 1 seed, `claude-sonnet-4-5` answer + judge, $7.74) was run to de-risk the adapters.

Result: the `longcontext` brute-force control scored J = **0.775**, while the keyless retrieval baselines scored J = **0.025** under the identical generate-then-judge protocol — a 0.75-point spread that token_f1 alone cannot provide (bm25 token_f1 floor is 0.046 here, yet J correctly credits its few right generations).

Vendors now measured on free tiers (polling ingest). Both hosted systems initially returned empty retrievals: Mem0's fact-extraction is *asynchronous* (a fixed post-ingest sleep was too short) and Zep's graph ingest needed time/credits to populate. Replacing the blind sleep with a `get_all` poll — waiting until the just-ingested memories are actually queryable, plus ingest-retry on transient errors — made all three measurable on the free tier. Letta additionally needed a local embedding endpoint (Ollama `nomic-embed-text`); once pointed at it, its archival memory ingested and deleted cleanly. **Mem0** J=0.475 (19/40, token_f1 0.144), **Letta** J=0.200 (8/40, token_f1 0.111), and **Zep** J=0.100 (4/40, token_f1 0.081), on the same p01 pilot (40 Q, 1 seed). The full ordering reads sensibly — longcontext 0.775 > Mem0 0.475 > Letta 0.200 > Zep 0.100 > retrieval floor 0.025 — real memory systems landing between brute-force long-context and the keyless floor, on real LoCoMo under one harness. **Only the full 5-seed/10-persona keyed run remains PENDING** (see also `docs/audit/VENDOR-PILOT-FINDING.md` and `MEMO-FREE-TIER-SUCCESS.md`). No number is invented; pending means pending.

Table 4: Standardized single-harness results (LoCoMo; Cat 1–4; J primary, token_f1 floor).

System	Config	J (s4-5)	J (robust.)	token_f1	single	multi	open	temporal	Status
bm25 (floor)	—	n/a	n/a	0.0596	0.0898	0.0291	0.0244	0.0179	**DONE ($0)**
tfidf (floor)	—	n/a	n/a	0.0520	0.0783	0.0242	0.0279	0.0146	**DONE ($0)**
longcontext	brute-force	**0.775** (pilot)	PENDING	0.201	—	—	—	—	**PILOT (40Q,1 seed)**
bm25/tfidf (gen)	—	*0.025*	—	*0.046/0.040*	—	—	—	—	*pilot floor*
mem0	vendor-blessed	**0.475** (pilot)	PENDING	0.144	—	—	—	—	**PILOT (free, 40Q, 1 seed)**
zep	vendor-blessed	**0.100** (pilot)	PENDING	0.081	—	—	—	—	**PILOT (free, 40Q, 1 seed)**
letta	vendor-blessed	**0.200** (pilot)	PENDING	0.111	—	—	—	—	**PILOT (free, 40Q, 1 seed)**

[*] Mem0/Zep could not be measured on free tiers (Mem0 async-extraction lag → empty retrievals; Zep graph 404 / credit exhaustion). Needs paid tiers + tuned processing waits. See §7.2 and `docs/audit/VENDOR-PILOT-FINDING.md`.
*Keyless floor rows carry token_f1 only (J = n/a): no generation, no generated answer to judge. The provenance-annotated claims table (Table 1) is **explicitly NOT** a column of this table.*

Table 5: Category-5 abstention diagnostic — PENDING keyed run.

System	Abstention rate (Cat-5, N=446)	Over-answer rate
bm25 (floor)	PENDING	PENDING
mem0	PENDING	PENDING
zep	PENDING	PENDING
letta	PENDING	PENDING

Never folded into J or token_f1; reported separately. Filled by the keyed run.

8 Threats to Validity / Limitations *[DRAFTED]*

- **Small dataset → wide variance.** LoCoMo is 10 conversations; we report ±std over ≥5 seeds.
- **Free-tier scale of the completed pilot.** The **completed p01 pilot** is a single-persona (p01), 1-seed run — it is a smoke test, not a leaderboard. The longcontext J=0.775 and floor J=0.025 are real but narrow-scope pilot results; headline J cells are reported only after the **full 5-seed, 10-persona keyed run** lands. Every preliminary cell is labeled accordingly.
- **Judge-model dependence.** Pinning one judge makes numbers internally consistent but not numerically identical to the `gpt-4o-mini`-era vendor cells; a second **robustness judge** with both J values reported per cell is the mitigation. **We never present a model-effect gap as an architecture gap.**
- **Single-provider answer + judge.** Both answer model and primary judge are Anthropic `claude-sonnet-4-5`. A systematic Claude-as-judge bias would shift all cells together; the cross-provider robustness judge is the intended check.
- **Vendor-config dependence.** A vendor declining sign-off is reported as such, with the default published config labeled.
- **What we did NOT verify (honesty ledger):**
  - [U] Judge/answer model behind Mem0's 92.5 and Zep's 75.14 — not stated on pages read; to be raised with vendors in the pre-disclosure window.
  - **[P]** The exact 84→58.44 inflation arithmetic — verified *as described* by Mem0, **not** independently re-executed (the audit's reproduction target).
  - [U] The Mem0 49.0 LongMemEval figure — found only in secondary aggregators; locate a primary run or drop it.
  - Mem0 paper Table-1 cell layout — re-read from the PDF (the auto-extractor mislabeled the 67.13 row as "single-hop"; it is the overall J).
  - Whether Zep replied inside issue #5 or only on its blog — comments not captured; re-read.

9 Ethics & Fair-Play *[DRAFTED]*

This audit's entire value is being **correct and fair**. (1) **Pre-disclosure:** no vendor learns its number first from a public post. (2) **Fairness cuts both ways:** we correct the record *for* Zep (the "84%" is not in its paper) as readily as we surface inconsistencies. (3) **Data ethics:** LoCoMo is CC BY-NC 4.0; we do not redistribute raw data. (4) **No legal claims:** this is a measurement audit, **NOT LEGAL ADVICE**, and makes no assertion of bad faith — the thesis is explicitly that the spread is a *measurement artifact*, not deception.

10 Reproducibility Statement & Release *[DRAFTED]*

Reproducibility statement. (1) **Public harness:** `eval/runner.py`, `eval/answer_gen.py`, `eval/judge.py`, `eval/metrics.py`, and every system adapter are released open-source. (2) **Pinned models:** answer + primary judge `claude-sonnet-4-5`, recorded in `manifest.audit`. (3) **Seeds:** ≥5 per cell; mean ± std. (4) **Frozen data:** `snap-research/locomo` @ `3eb6f2c`; clone + commit-pin instructions supplied. (5) **Deterministic outputs:** `-tag` fixes the result filename; re-invoking resumes without re-spending completed cells. (6) **Exact commands:** the keyless floor (Table 3) reproduces today at $0.00.

11 Conclusion *[DRAFTED]*

The LoCoMo dispute is a **measurement artifact, not an architecture verdict.** There is no single LoCoMo score because every headline number was produced under a different protocol, often by the competitor of the system being scored, and several famous comparisons are cross-source splices. Named and held constant, the 84/58.44/75.14/92.5 spread resolves into four methodological forks. A single frozen, vendor-sign-off, fully-reproducible protocol — run on the real data, with a pre-disclosure fair-play window — is the durable, citable contribution. The keyless real-data floor is reported now; the standardized vendor numbers are the empirical payoff once the keyed run lands.

Venue plan. **Now:** arXiv preprint — establishes the citable artifact and timestamps the provenance correction and the frozen protocol. **After the keyed run:** ACL/EMNLP 2026 resource/benchmark or short-paper track, or NeurIPS 2026 Datasets & Benchmarks / Evaluation & Disclosure. **Companion:** open-source harness + project page.

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